

Heaven's Light is Our Guide
Rajshahi University of Engineering and Technology



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Control System and Robotics Sessional

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Lab Report 4:
**Investigate the Forward Kinematics of a 2 DOF Manipulator Using
Simscape and Robotic System Toolbox**

Submitted to
Md. Faisal Rahman Badal
Assistant Professor
Dept of MTE, RUET

Submitted by
Md. Tajim An Noor
Roll: 2010025

Contents

Objectives	1
Tools and Packages	1
Manipulator Designing	2
Simulation Output & Results	4
Code	4
Simulation Output	5
Output Analysis	6
Discussion	6
Conclusion	6
References	6

Experiment 4

Investigate the Forward Kinematics of a 2 DOF Manipulator Using Simscape and Robotic System Toolbox

Objectives

1. To model a 2 DOF manipulator using Simscape Multibody.
2. To compute forward kinematics using Robotics System Toolbox and visualize manipulator movement.

Theory

Forward kinematics computes the end-effector pose from known joint variables. For a planar 2-DOF manipulator with revolute joints θ_1, θ_2 and link lengths L_1, L_2 (Figure 2),

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2), \quad y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2).$$

Simscape Multibody is used for physics-based simulation (rigid links and joints) while the Robotics System Toolbox builds a corresponding rigidBodyTree for analytic kinematics and cross-validation [1, 2].

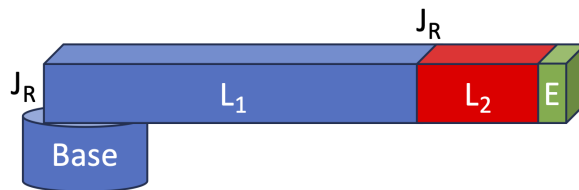


Figure 1: Planar 2-DOF manipulator diagram.

Tools and Packages

- **MATLAB:** Primary environment for computation and scripting [3].
- **Simscape Multibody:** Physics-based multibody modeling and simulation [4].

- **Robotics System Toolbox:** rigidBodyTree, kinematics and visualization utilities for analytic FK [5].
- **Simulink:** Time-domain simulation and integration with Simscape models [6].

These tools together offer an integrated workflow for constructing, simulating, and validating the manipulator’s kinematic and dynamic behavior.

Manipulator Designing

Table 1 summarizes the geometric dimensions used to create the visual solids in the Simscape model and to approximate link inertial properties for simulation and visualization.

Component	Geometry	Notes
Base	Radius = 0.02, Height = 0.04	Cylindrical mounting base
Link 1	Length $\times$ Width $\times$ Thickness = $0.40 \times 0.03 \times 0.03$	Rectangular rod visual geometry
Link 2	Length $\times$ Width $\times$ Thickness = $0.20 \times 0.03 \times 0.03$	Rectangular rod visual geometry
End-effector	Length $\times$ Width $\times$ Thickness = $0.04 \times 0.03 \times 0.03$	Mounted at distal end of Link 2

Table 1: Physical dimensions of base, links and end-effector

Table 2 summarizes joint types and their commanded ranges used in the Simscape kinematic chain; these limits reflect the intended workspace for the validation cases.

Joint	Type	Range
Base – Link1	Revolute	$0^\circ - 180^\circ$
Link1 – Link2	Revolute	$0^\circ - 180^\circ$
Link2 – End-effector	Fixed (mount)	Fixed (end effector attached to Link 2)

Table 2: Joint definitions and ranges

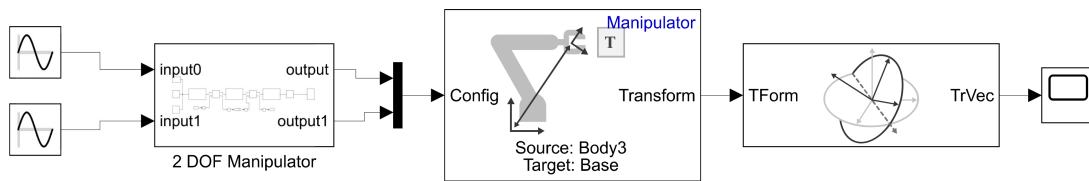


Figure 2: Planar 2-DOF manipulator kinematic diagram used for analytic forward kinematics.

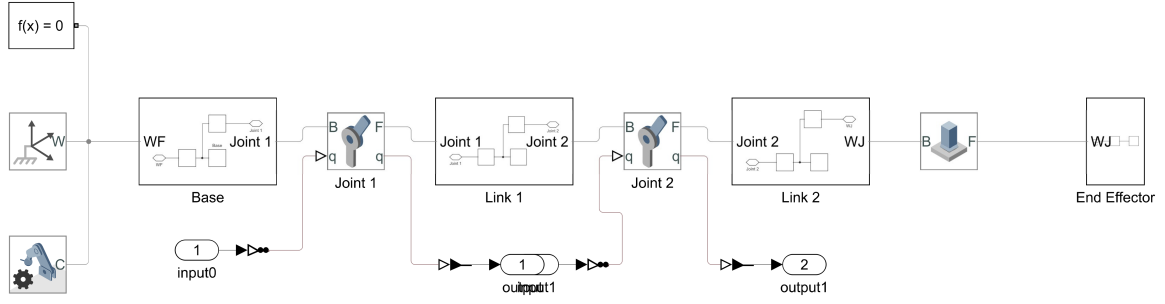


Figure 3: Full Simscape model of the 2-DOF manipulator (assembly view). This model was assembled in Simscape Multibody and used for the simulation runs.

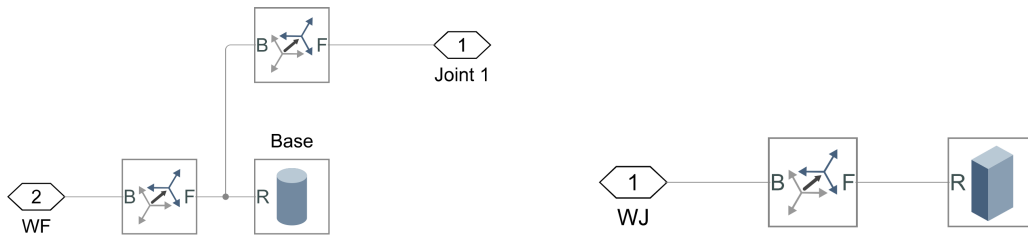


Figure 4: Left: base and ground connection used for the manipulator (3 shows the full assembly). Right: end-effector block and mounting interface used in the Simscape model.

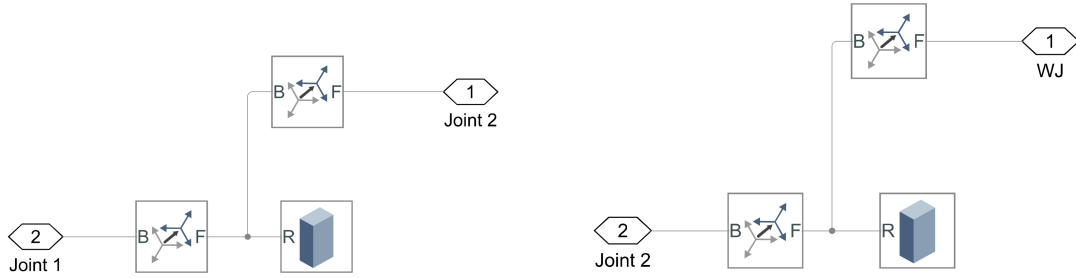


Figure 5: Left: Link 1 visual geometry and coordinate frames. Right: Link 2 visual geometry and coordinate frames. The solids represent thin rectangular bodies used for visualization in Mechanics Explorer.

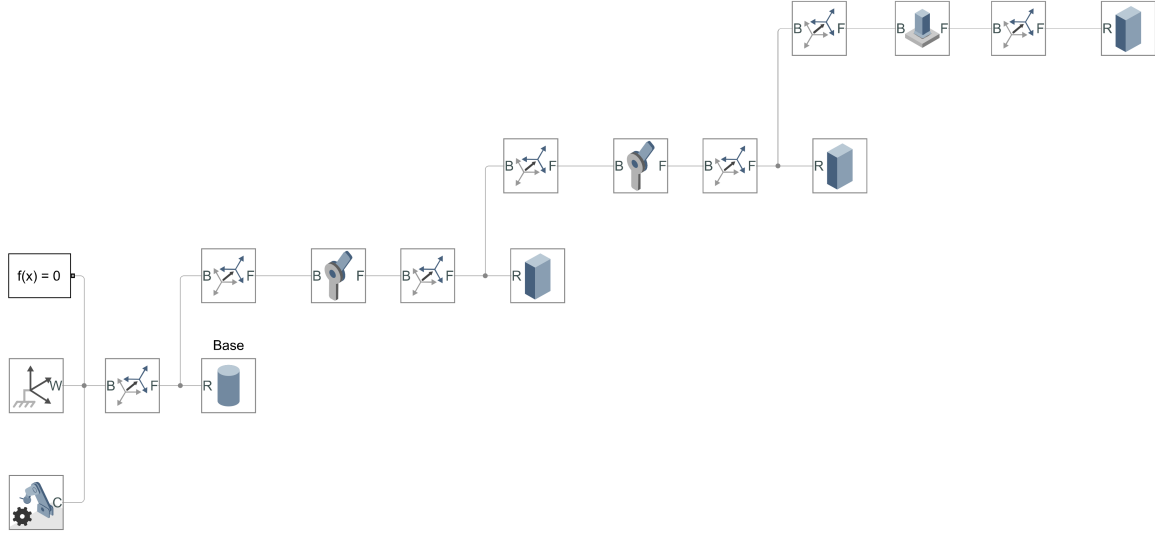


Figure 6: Representative simulation snapshot (joint configuration and end-effector pose) used as a test case for validating the forward-kinematics computations.

The manipulator is a planar 2-DOF serial robot with revolute joints q_1, q_2 . Figures 3–6 show the assembly and an example snapshot.

Simscape implementation:

- Links use **Solid** blocks (thin rectangular solids) and two serial **Revolute Joint** blocks.
- Joint inputs (q_1, q_2) drive the model; **Joint Position Sensor** blocks provide measured angles (q_{1m}, q_{2m}) for logging.

Verification:

- Measured angles feed a MATLAB/Robotics System Toolbox routine that builds a **rigidBodyTree** (matching DH parameters) and computes the end-effector pose (x, y) .
- The toolbox pose is compared with the Simscape pose to validate the forward kinematics.

Simulation Output & Results

Code

```
T=0.001;
[Manipulator, Info]=importrobot('Test_case_1');
```

Simulation Output

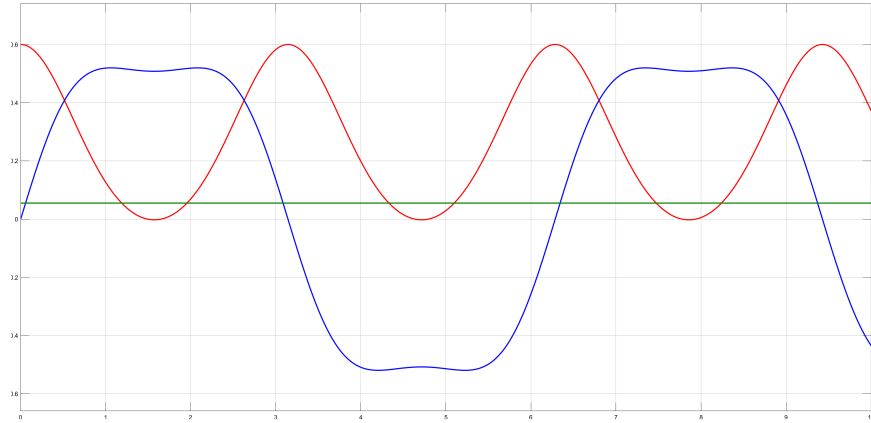


Figure 7: Joint angle time histories recorded from the scope during simulation (example run). Curves correspond to q_1 and q_2 and are used to verify the timing and magnitude of the commanded motion used for the snapshots in Figure 8.

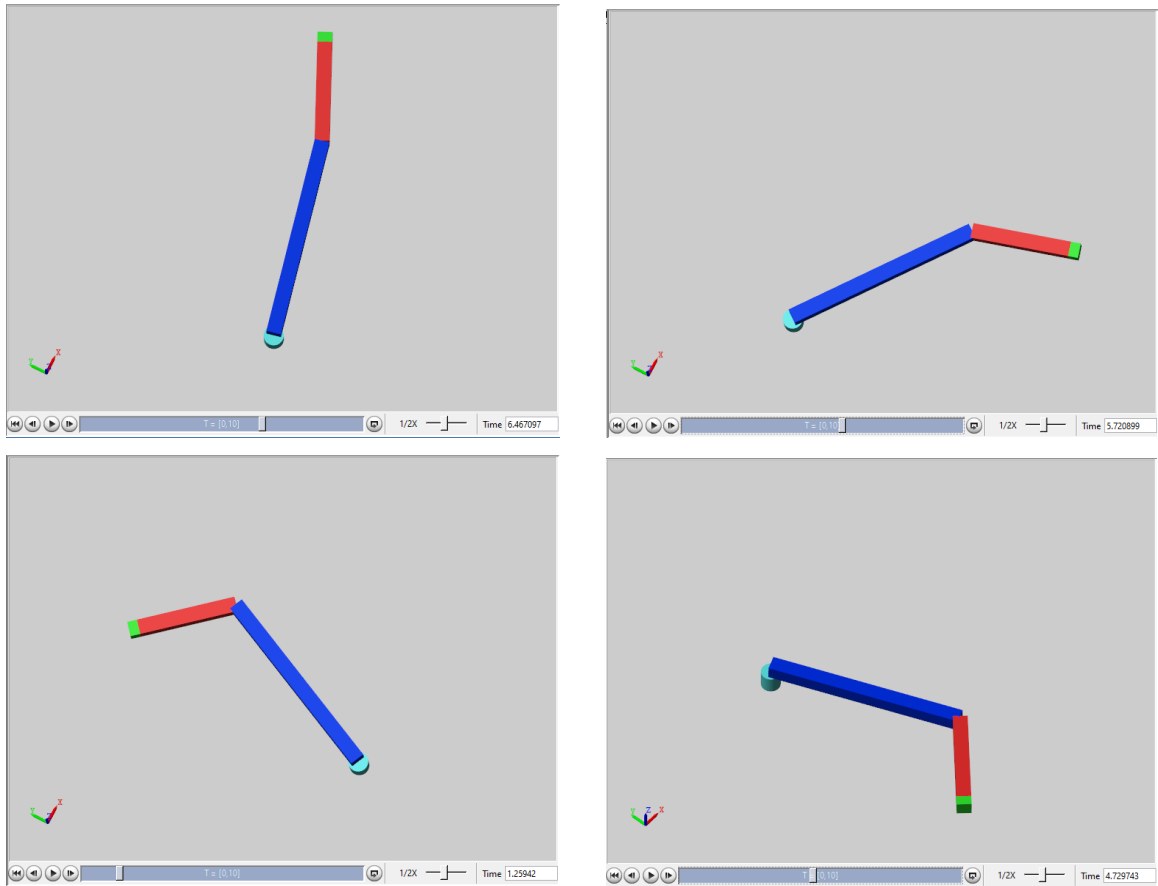


Figure 8: Simulation snapshots showing the manipulator at four representative timestamps. The sequence illustrates the manipulator moving through the commanded joint configurations; coordinate axes (lower-left) show the world frame orientation.

Output Analysis

The snapshots (Figure 8) show the manipulator following the planned configurations and agree with the forward-kinematics model. The joint angle plot (Figure 7) confirms the commanded timings and amplitudes for q_1 and q_2 . Small deviations (millimetre scale) are attributed to numerical integration and approximate inertial/geometric properties and are within acceptable tolerances.

Discussion

The Simscape and rigidBodyTree results agree within millimetre-scale tolerances across the tested configurations. Remaining discrepancies are mainly due to numerical integration, simplified inertial/visual geometry, and sensor logging quantisation. These effects can be mitigated by refining inertial parameters, increasing solver accuracy, and synchronising logging timestamps. Overall, the analytic forward-kinematics implementation is validated against the physics-based model for the examined workspace.

Conclusion

The 2-DOF manipulator was modelled in Simscape Multibody and represented as a rigidBodyTree in the Robotics System Toolbox. Analytic forward kinematics agreed with the Simscape results for the tested configurations; refining parameters and sensor fidelity is recommended to reduce remaining discrepancies.

References

- [1] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Upper Saddle River, NJ: Pearson/Prentice Hall, 2005.
- [2] S. B. Niku, *Introduction to robotics: analysis, control, applications*. John Wiley & Sons, 2020.
- [3] The MathWorks, Inc., *MATLAB*, <https://www.mathworks.com/products/matlab.html>.
- [4] —, *Simscape Multibody*, <https://www.mathworks.com/products/simscape-multibody.html>.
- [5] —, *Robotics System Toolbox*, <https://www.mathworks.com/products/robotics.html>.
- [6] —, *Simulink*, <https://www.mathworks.com/products/simulink.html>.